Adroit Technologies Protocol Driver

Name	Hitachi Ethernet TCP/IP
DLL	HITACHIE

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Contents

1. Overview

Adroit supports communication with the following Hitachi PLC CPU's - EH150, H200, H250, H252A/B/C, H300, H302, H700, H702, H1002, H2000, H2002 and H4010 via a dedicated Ethernet module.

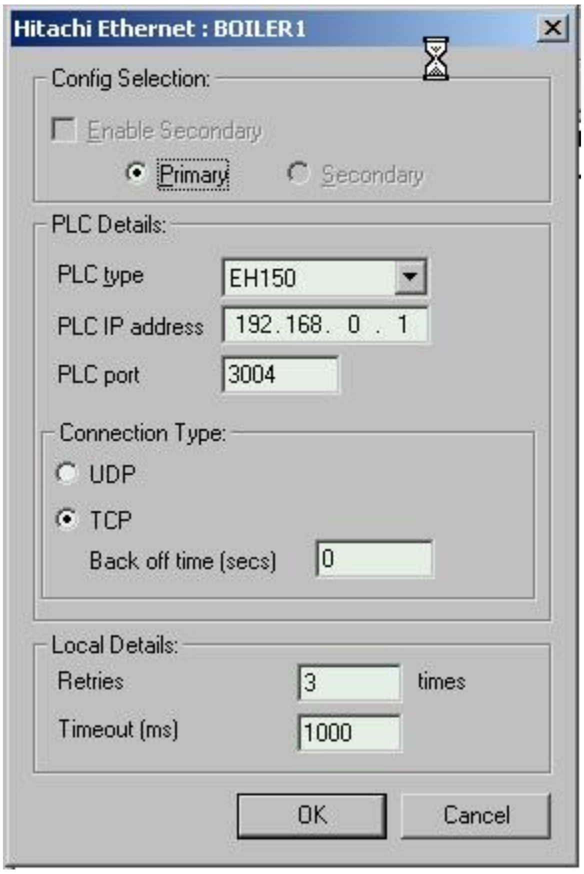
The three Ethernet modules that have been tested against this driver are the EH ETH (EH150), HLAN (H200-H252C) and the LAN ETH2 (H300-H4010).

2. Wiring

Standard Ethernet interconnection (10BASE T/E).

3. Device Configuration

3.1. Device Configuration



3.1.1. Config Selection [not implemented]

Enable Secondary – (feature disabled) Enables secondary configuration for dual redundant or backup PLC's.

Primary / Secondary – (feature disabled) Select the primary or secondary button depending on which connection you wish to configure.

3.1.2. PLC Details

The maximum amount of data per scan is shown here :

Data type	Max number of read elements	Max number of write elements
Words	120 words	100 words
Bits	240 bits	200 bits

PLC IP Address – The TCP/IP Address of the PLC.

PLC Port – This is the TCP or UDP port, typically 3004 through 3007.

3.1.3. Connection Type

Connection Type – Choose whether the protocol driver should implement the TCP or UDP protocol.

Back Off Time – As of rev 10c of the driver, this backoff time is unused due to a strange anomaly found with multiple Hitachi PLC’s on an Ethernet network whereby the time taken to establish a connection to a PLC is dependent on the number of PLC’s on the network.
Therefore, the “back off time” time has been replaced by a hardcoded timing/backoff mechanism for reconnecting to PLC’s. The hardcoded algorithm is dependent on the number of configured devices in groups of 10. The value is 10 seconds for the first 10 devices + 20 seconds for each multiple of 10 devices or part thereof thereafter. For example, 26 PLC's will take up to 50 (10+20+20) seconds to reconnect.

The following timing table refers:

No of PLC’s	Time to establish connection
1-10	10 secs
11-20	30 secs
21-30	50 secs
31-40	60 secs

The reconnection takes place on a background thread.

3.1.4. Local Details

Retry – The number of times the driver must retry a Transmit/Receive transaction before returning an indication of failure.

Timeout – The time in milliseconds to wait between each communication retry. Please read the Back Off Time description in this section. For the same reason, experience has shown that the timeout value used here needs to take into consideration the number of PLC’s on the network. No definitive relationship has been established, but it has been found in practice that a normal timeout value of 1 second which would normally work on a network of 10 PLC’s or less, will not work on a network of 11 PLC’s or more. It has been found that on a network of 21 to 30 PLC’s, a timeout value of 35 seconds was required.

4. Supported Addresses

Hitachi Ethernet PLC data address types supported by the Hitachi Ethernet Adroit Protocol driver. Please refer to the relevant Hitachi CPU manuals for ranges, and or check the driver address options prompt.

Hitachi PLC data address types supported by the Hitachi Adroit Protocol driver:

Device	PLC Data Type	Boolean	Integer	Real
R Area	Digital	R0000	-	-
M Area	Digital	M0000	-	-
L Area	Digital	L0000	-	-
X Area	Digital	X0000	-	-
Y Area	Digital	Y0000	-	-
WR Area	16-bit WORD Reg	-	WR0000	WR0000
	16-bit SHORT Reg	-	WR0000 I	WR0000 I
WM Area	16-bit WORD Reg	-	WM0000	WM0000
	16-bit SHORT Reg	-	WM0000 I	WM0000 I
WX Area	16-bit WORD Reg	-	WX0000	WX0000
	16-bit SHORT Reg	-	WX0000 I	WX0000 I
WY Area	16-bit WORD Reg	-	WY0000	WY0000
	16-bit SHORT Reg	-	WY0000 I	WY0000 I
WL Area	16-bit WORD Reg	-	WL0000	WL0000
	16-bit SHORT Reg	-	WL0000 I	WL0000 I
DR Area	32-bit DWORD Reg	-	WR0000	WR0000
	32-bit LONG Reg	-	WR0000 L	WR0000 L
	32-bit FLOAT Reg	-	-	DM0000 R
DM Area	32-bit DWORD Reg	-	DM0000	DM0000
	32-bit LONG Reg	-	DL0000 L	DL0000 L
	32-bit FLOAT Reg	-	-	DL0000 R
DL Area	32-bit DWORD Reg	-	DM0000	DM0000
	32-bit LONG Reg	-	DL0000 L	DL0000 L
	32-bit FLOAT Reg	-	-	DL0000 R

PLEASE NOTE

Write functionality to the X and WX area (Status register area) have been disabled, due to it being an input register.

Floats are defined as IEEE-754.

5. Protocol Definition and Considerations

5.1. Hitachi Task Command Header Structure

Prefix structure for all Read/ Write task messages.

Hitachi packet header: (The numbers are in **hexadecimal** unless otherwise noted)

Start Code	Sequence no.	L	U	M	P	Task Code	Task Code Parameters
00	12	FF	FF	00	00	xx	xx

Start Code:
The start code indicates that we are issuing a new command when it is set to 0x00.

Sequence no:
The sequence no. can be anything from 0x01 to 0xFF. This number will be the same in the reply packet. This would be used to distinguish the correct reply packet from numerous incoming packets on a busy network.

LUMP Address: (H Series Network Address):
L: Loop No.
The loop no. for a CPU Link module installed. In case a CPU link module is not installed or a station to link is not accessed, this number is set to 0xFF. For this protocol driver, the number is always set to 0xFF.
U: Unit No.
Station no. of a CPU Link module installed. In case a CPU link module is not installed or a station to link is not accessed, this number is set to 0xFF. For this protocol driver, the number is always set to 0xFF.
M: Module number.
If set to 0x00, this number indicates a normal CPU module. It is always set to 0x00 in this protocol driver. An error in the communication will result in the response packet having a different module and/or port number, to indicate the origin of the error. This will result in a communications failure in the protocol driver, after which the sequence of port retry and the back off time will be implemented.
P: Port number.
If set to 0x00 indicates that it is a normal CPU module port.

Task code and task code parameters:
The specific task to be performed, along with its parameters, can be set here. The task code is 1 byte long (for ex. 0x0A).

5.2. Hitachi Continuous N points monitor (READ) function

Command Format:

Task Code	I/O Code.	I/O Address.	No. of items to read
A0	01	0007ff	01

Response Format:

Execution status	Task Code	Monitor data
00	A0	Xxxx

Command Code Parameters:

Task Code:

0xA0 is the task code reading data from the PLC.

I/O Code:

The I/O Code indicates the memory area to read from. All the available memory area codes are listed under the section 'I/O memory area codes' later in this document.

I/O Address:

The I/O Address is the memory address from where the reading of the specified number of items will commence. It is a 6-digit hexadecimal number (0x000000).

No. of items to read:

The number of items (bits, words or dwords) to be read.

Response Code Parameters:

Execution status:

If the execution status is set to 0x00, the task code executed correctly in the PLC.

Task Code:

The task code that was performed is returned in the response packet.

Monitor data:

The data that was requested.

5.3. Hitachi Continuous N forced set/reset (WRITE) function

Command Format:

Task Code	I/O Code.	I/O Address.	No. of items to write	Data to write
A2	01	0007ff	01	xxxx

Response Format:

Execution status	Task Code
00	A2

Command Code Parameters:

Task Code:

0xA2 is the task code writing data to the PLC.

I/O Code:

The I/O Code indicates the memory area to write to. All the available memory area codes are listed under the section 'I/O memory area codes' later in this document.

I/O Address:

The I/O Address is the memory address from where the writing of the specified number of items will commence. It is a 6-digit hexadecimal number (0x000000).

No. of items to read:

The number of items (bits, words or dwords) to be written.

Data to write:

The data that must be written to the specified memory area.

Response Code Parameters:

Execution status:

If the execution status is set to 0x00, the task code executed correctly in the PLC.

Task Code:

The task code that was performed is returned in the response packet.

Monitor data:

The data that was requested.

5.4. I/O Memory Area Codes

Area	Data type	Code
R	Bit	02
WR	Word	0A
DR	Dword	10
L	Bit	03
WL	Word	0B
DL	Dword	11
M	Bit	04
WM	Word	0C
DM	Dword	12
X	Bit	00
WX	Word	08
Y	Bit	01
WY	Word	09

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback (adroittechnologiesautomation.com)

6.3. Troubleshooting

When Addressing the X and Y regions as Booleans the following indicates how:
X/Y 0000 displayed as X ABCD means the following:
A - The base number where the input or output card is plugged in.
B - The Slot Number where the input or output card is plugged in.
CD - The Input or Output Address of that card ranging from 00 to 3F.
Example: X 0100 means you require input 0 from the input card plugged into slot 1 of base 0.

When Addressing the X and Y regions as Integers/Reals the following indicates how:
WX/WY 0000 displayed as WX ABCD means the following:
A - The base number where the input or output card is plugged in.
B - The Slot Number where the input or output card is plugged in.
CD - The Input or Output Address of that card ranging from 00 to 30 depending on your input/output card.
Example: WX 0110 means you require input word 2(Bits 16 to 31) from the input card plugged into slot 1 of base 0.

The addressing of WX and WY only occur on multiples of 16, so that means that we can only make CD equal to 00,10,20,30 where 00 will get the whole word from input 0 to 15 and 10 will get the whole word containing input 16 to 31 etc.

If your input card has 16 inputs you can only scan CD = 00.
If your input card has 32 inputs you can only scan CD = 00 and CD = 10.
If your input card has 48 inputs you can only scan CD = 00 and CD = 10 and CD = 20.
If your input card has 64 inputs you can only scan CD = 00 and CD = 10 and CD = 20 and CD = 30.

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	25-Jan-2023	K. Robinson	Changes to format, updated image resources. Added Appendix B DCOM Hardening. Removed Windows XP section	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

7.2. DLL Revision

Rev.	Date	Changes
1	19-Jan-2005	First release notes! Beta Release Available
	14-Jan-2005	Post commissioning updates
	11-Oct-2005	CDB: Back-off mechanism paragraph moved lower down and revised.
	29-Aug-2006	Reworked many aspects of the driver , corrected problems with i/o formatting.
	25-Aug-2006	MGL: Added notes about configuration of backoff time and timeout value as they pertain to rev 10c of the driver
	09-Feb-2007	CCS: Got UDP working as it was not working at all and changed the way we address WX, WY, X, and Y addresses. In TROUBLESHOOTING you will find a thorough explanation on these Addressing.